Fizika yechimlari

1) ⑧ball $y = 5x^2$ $\begin{cases} x = 2t \\ y = 8t \end{cases} \Rightarrow \boxed{t = \frac{x}{2}}$ ①

$\boxed{y = 8\cdot t = 8\cdot \frac{x}{2} = 4x}$ ①

Ikkita zarrachaning uchrashish sharti:

$\boxed{\begin{cases} y = 5x^2 \\ y = 4x \end{cases}}$ ① Shu tenglamalar sistemasi yechimga ega bōlsa, zarrachalar uchrashadi

$\boxed{5x^2 = 4x}$ ikkita tenglamani bir-biriga tenglab yuboramiz va x uchun tenglamani yechamiz

$x \neq 0$ $\boxed{5x = 4}$

$\boxed{x = \frac{4}{5}}$ ① yechim bor, demak zarrachalar uchrashadi. ①

$\begin{cases} x = 0.8 \\ \boxed{y = 4x = \frac{16}{5} = 3.2} \end{cases}$ ① koordinatalarga ega bōlgan nuqtada

Vaqtning $\boxed{t = \frac{x}{2} = 0.4}$ ① momentida uchrashadi.

Xususiy hol: Agar $x = 0$ bōlsa, ya'ni

$5x^2 - 4x = 0$ $x(5x - 4) = 0$ tenglamaning

$x = 0$ yechimi ham bor. Bunda $y = 0$ bōladi

Ya'ni ikkala zarracha ham koordinata boshida joylashgan bōlsa, ular (0,0) nuqtada uchrashgan bōladi. ⑦

2) Berilgan:

$t = 96\ h = 345600\ s$

$N = 64$

$h = 230\ km = 2{,}3\cdot10^5\ m$

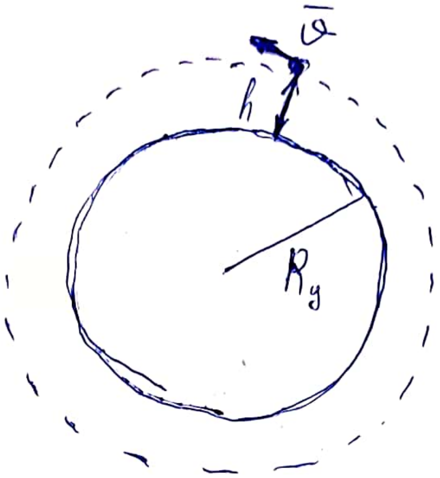
$R_y = 6370\ km = 63{,}7\cdot10^5\ m$

$\pi = 3$

$\bar{v} = ?$

$S_u = \ell\cdot N$

$\bar{v} = \frac{S_u}{t_u} = \frac{2\pi(R+h)N}{t_u}$



Berilgan kattaliklarni tōgi yozsa va chizmani tōgri chizsa 2 ball

Yechish:

$$\begin{cases} T = \frac{t}{N} \\ \text{yoki}\quad T = \frac{2\pi(R+h)}{\bar{v}} \end{cases} \Rightarrow \frac{t}{N} = \frac{2\pi(R+h)}{v} \Rightarrow \bar{v} = \frac{2\pi N(R+h)}{t}$$

Masalaga tegshli formulalarni yozsa va ulardan foydalangan holda orbita bōylab ōrtacha tezligini hisoblash formulasini keltrib chiqarsa jami 3,5 ball.

$$\bar{v} = \frac{2\cdot3(63{,}7\cdot10^5+2{,}3\cdot10^5)\cdot64}{345600} = \frac{6\cdot64\cdot66\cdot10^5}{345600} = 7{,}(3)\cdot10^3\ \frac{m}{s}$$

Javob: $\bar{v} = 7{,}(3)\cdot10^3\ \frac{m}{s} = 26{,}4\cdot10^3\ \frac{km}{h}$

tezlikni hisoblash formulasig berilgan kattaliklarni qōyib tōgri hisob kitob qilsa jami 2,5 ball

Berilgan masalani tōliq yechsa Jami: 8 ball

N3.

$F_2 = 20\,N$

$F_1 = 5\,N$

$P - ?$

$$\begin{cases} P \cdot x = F_1 \cdot y & (2) \\ P \cdot y = F_2 x & (2) \end{cases}$$

$$\frac{P}{F_2} = \frac{F_1}{P}$$

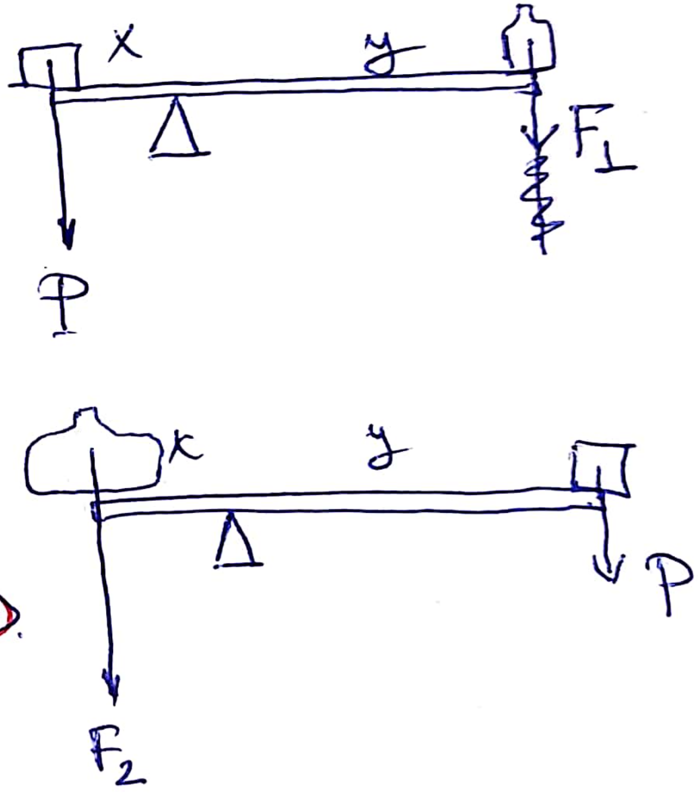
$$P^2 = F_1 F_2 \quad (2)$$

$$P = \sqrt{F_1 F_2} = \sqrt{20 \cdot 5} = 10\,N \quad (2)$$

$$m = \frac{P}{g} = 1\,kg$$

Jami: 10 ball

(Chizmaga 2 ball)



N4.

Moshing gildiraklariga tasir qilayotgan F_{ish} oldinga yo'nalgan (2)

$\alpha = 37^\circ$

$a = 0{,}4\ m/s^2$

$\sin(37^\circ) = 0{,}6$

$\cos(37^\circ) = 0{,}8$

$\mu_{min} - ?$

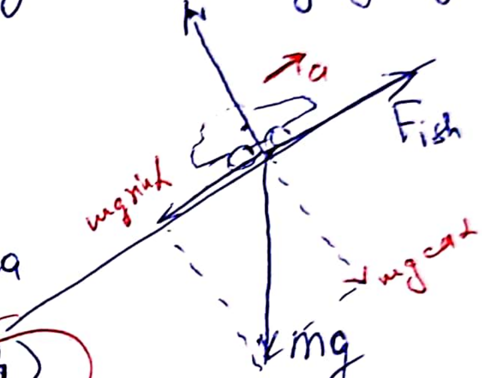
$$\begin{cases} F_T - mg\sin\alpha = ma \\ F_{ish,m} \geq F_T \end{cases}$$

μ ning minimal qiymatida

$F_{ish,} - mg\sin\alpha = ma$ (3)

$\mu mg\cos\alpha - mg\sin\alpha = ma$

$$\mu = \frac{a + g\sin\alpha}{g\cos\alpha} \quad (4)$$



(Chizmaga (1 ball))

Hisoblash: $\mu = \dfrac{0{,}4 + 10 \cdot 0{,}6}{10 \cdot 0{,}8} = \dfrac{6{,}4}{8} = 0{,}8$ (2)

Jami: 12 ball

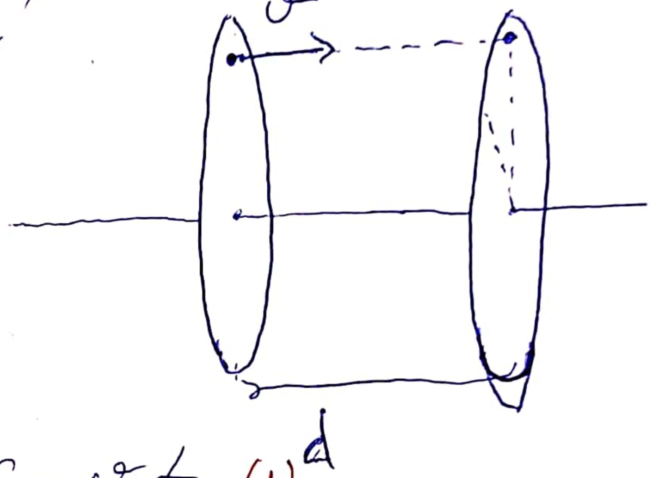
5-masala | Berilgan:

$d = 0{,}5\ m$ (1)

$n = 1600\ \frac{ay}{min} = \frac{1600}{60} = \frac{80}{3}\ \frac{ay}{s}$

$\Delta\varphi = 12^\circ = \frac{\pi}{15}$ (1)

$\vartheta - ?$



Ōqning harakati: $S = \vartheta t$ (1)

Diskning aylanma harakati: $\varphi = \omega t$ (3)

$\omega = 2\pi\nu$ (1)

Ōqning ikki disk orasidagi harakat vaqti bilan, diskning shu vaqt ichida burilishini ~~Vaqtlarini~~ tenglash orqali masalani yechamiz. (4)

$d = \vartheta \cdot t \qquad \Delta\varphi = 2\pi\nu t$

$\frac{d}{\vartheta} = \frac{\Delta\varphi}{2\pi\nu} \rightarrow \boxed{\vartheta = d \cdot \frac{2\pi\nu}{\Delta\varphi}}$ (2)

$\vartheta = 0{,}5 \cdot \frac{2 \cdot \cancel{\pi} \cdot 80 \cdot 15}{3 \cdot \cancel{\pi}} = 400\ m/s$ (1)

Javob: $\vartheta = 400\ m/s$